DATA ANALYTICS

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Covering: Data Analytics | Data Analysis Process | Data Cleaning | Statistics | Visualization

Chapter 1: What Is Data Analytics?

Definition of Data Analytics
Data analytics is the process of analyzing raw data in order to draw out meaningful, actionable insights. It is a form of business intelligence that enables companies and organizations to make smart decisions based on what the data is telling them.

Simple Explanation

Think of data analytics like reading a story hidden inside numbers and facts.
A company collects a lot of data every day - from sales, customers, website visits, etc.
Data analytics helps us read that data, find patterns, and make smart business decisions.
It covers: collecting raw data, preparing it, analyzing it, and sharing the key findings.

Real-Life Example

Example: Public Transport Planning
Imagine you work for Dhaka BRTA (like MTA in New York). A big cricket match is coming to Mirpur Stadium and thousands of fans will travel. How do you plan extra buses and trains? You use data analytics! You analyze data from past big events - how many people traveled, which routes were busiest, what times were peak hours. Using this data, you predict what will happen during the cricket match and plan accordingly. This replaces guesswork with data-driven decisions.

Important Notes for Exam

- Data analytics = analyzing raw data to get useful insights
- It is a type of business intelligence
- It helps replace guesswork with facts
- It includes: data collection, preparation, analysis, and storytelling (sharing results)

Chapter 2: What Does a Data Analyst Do?

Role of a Data Analyst
A data analyst's job is to turn raw data into meaningful insights. They start with a specific problem or question, then collect, clean, analyze, and present data to answer that question.

Key Steps in a Data Analyst's Work

1. Define the Question - Identify the business problem to solve
2. Collect the Data - Gather relevant data from various sources
3. Clean the Data - Remove errors, duplicates, and irrelevant data
4. Analyze the Data - Use statistical tools to find patterns
5. Create Visualizations - Make charts, graphs, and dashboards
6. Share Your Findings - Present insights to decision-makers

Example: Business Questions a Data Analyst Might Solve
Question 1: 'Why did we lose so many customers last quarter?' - A data analyst would collect customer data, find patterns in when and why customers left, and provide a clear answer. Question 2: 'Why are patients dropping out of therapy at the halfway mark?' - A data analyst would study patient records, look for common factors among dropouts, and identify the root cause.

Chapter 3: The Data Analysis Process (7 Steps)

Overview
The data analysis process is a step-by-step method that a data analyst follows to solve a business problem using data. There are 7 main steps in this process.

Step 1: Defining the Question

What is it?
The first step is to clearly define what problem you want to solve. This is also called the 'problem statement'. You form a hypothesis and figure out how to test it.

This step is mostly about understanding the business and asking the RIGHT question.
A surface-level question like 'Why are we losing customers?' may not get to the real problem.
A better question: 'Which factors are negatively impacting customer experience?' or 'How can we boost customer retention while minimizing costs?'
You need business knowledge + logical thinking to frame the problem correctly.
Tools: KPI Dashboards (e.g., Grafana, Databox, DashThis)

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Tools used here: KPI Dashboards such as Grafana, Databox, and DashThis.

Example: TopNotch Learning Company
TopNotch Learning makes custom training software. They are great at winning new clients but have low repeat business. Instead of asking 'Why are we losing customers?', a good analyst asks: 'How can we boost customer retention while minimizing costs?' This is a more specific and actionable question.

Step 2: Collecting the Data

What is it?
Once the objective is set, you create a strategy to collect the right data. You decide what type of data you need and where to get it from.

Types of Data

Type	What It Is	Example
First-Party Data	Data you collect directly from your own customers or operations	Bkash transaction records, customer surveys, CRM data
Second-Party Data	First-party data of another organization, shared or purchased	A partner app's user activity data, shipping data from a logistics company
Third-Party Data	Data collected from many sources by an outside organization	Kaggle datasets, Google Dataset Search, government portals like data.gov

Quantitative vs Qualitative Data

Type	Definition	Example
Quantitative (Numeric)	Data expressed as numbers	Sales figures: 10,000 BDT; Number of users: 5000
Qualitative (Descriptive)	Data expressed as descriptions or text	Customer reviews, interview responses, survey answers

Free Data Sources (Third-Party)

- Google Dataset Search
- Kaggle (www.kaggle.com)
- Datahub.io
- UCI Machine Learning Repository
- NASA Earth Data
- USA Government: data.gov

Tools to Collect Data

- Data Management Platform (DMP): Software that identifies, aggregates, and manages data from multiple sources
- Enterprise DMPs: Salesforce DMP, SAS, Xplenty
- Open-source DMPs: Pimcore, D:Swarm

Step 3: Cleaning the Data

What is it?
Data cleaning (also called 'scrubbing') means preparing the collected data for analysis by removing errors, inconsistencies, and irrelevant information. This ensures high-quality data.

Key Data Cleaning Tasks

Task	What It Means	Why It Matters
Remove errors, duplicates, outliers	Delete repeated or incorrect entries	Bad data leads to wrong analysis
Remove unwanted data points	Discard data that is irrelevant to your question	Keeps the dataset focused and clean
Bring structure to data	Fix typos, layout issues, capitalization	Makes data easier to process
Fill in gaps	Identify and fill in missing important data	Incomplete data gives incomplete results

8 Steps of Data Cleaning

7. Get rid of unwanted observations - Remove irrelevant data points that do not match your analysis goal. Example: For vegetarian eating habits analysis, remove all meat-related records.
8. Fix structural errors - Fix typos and inconsistent capitalization from manual data entry. Example: 'Iron' and 'iron' should be standardized to one format.
9. Standardize your data - Make sure every entry follows the same rules. Use same units (all kg or all lbs), same date format, same capitalization throughout.
10. Remove unwanted outliers - Outliers are data points very far from the average. Only remove them if you can prove they are errors.
11. Fix contradictory data errors - Catch cross-set errors where records are inconsistent with each other. Example: Total race time not matching the sum of individual race times.
12. Fix type conversion and syntax errors - Ensure numbers are stored as numbers, dates as date objects, text as text - not mixed up.
13. Deal with missing data - Three options: (1) Remove the record, (2) Impute/estimate the missing value, (3) Flag it as 'missing' so analysis still accounts for it.
14. Validate your dataset - Final check: make sure all cleaning steps are complete and data is ready for analysis.

Benefits of Data Cleaning

- Staying organized: Clean data is easier to store and retrieve securely
- Avoiding mistakes: Marketing teams using clean databases send personalized messages without errors
- Improving productivity: Teams waste less time searching through messy data
- Avoiding unnecessary costs: Catching errors early prevents expensive mistakes later
- Improved data mapping: Clean data makes it easier to build new systems and applications

What is an Outlier?

Outlier
An outlier is a single data point that goes far outside the average value of a group of statistics. It is a value that is very different from all other values in the dataset.

Example: Outlier in Student Grades
10 students took a test. Their grades were: 85, 85, 90, 22, 92, 95, 91, 93, 87, 89. Most students scored between 85 and 95. But Student 4 scored only 22. This score is very far from the rest - it is an OUTLIER. It could be due to an error, illness, or the student not attending class. We must investigate before removing it.

Tools for Data Cleaning

- OpenRefine - Free, open-source, good for basic cleaning
- Python (Pandas library) - Best for large datasets and heavy scrubbing
- R packages - Good for statistical cleaning
- Data Ladder - Enterprise-grade data matching and cleaning tool

Data Quality

Definition
Data quality measures how well a dataset is suited for its intended purpose. High quality data is accurate, complete, consistent, valid, uniform, and relevant.

Quality Dimension	Definition	Example
Accuracy	Is the data correct?	A customer's phone number must be a real, working number
Completeness	Is all necessary data present?	A customer contact list must have all names if the goal is alphabetical sorting
Consistency	Does the data match across all sources?	A patient cannot have two different dates of birth in two systems
Validity	Does the data follow the required format and rules?	A birth year of 1700 is invalid - it falls outside the accepted range
Uniformity	Are all values in the same unit or format?	Weight data must all be in kg OR all in lbs - not mixed
Relevance (Timeliness)	Is the data available when needed and fit for the task?	Board meeting needs current quarter profits, not last quarter's figures

Step 4: Analyzing the Data

What is it?
After cleaning, you analyze the data to find patterns, trends, and answers to your question. The type of analysis depends on your goal.

4 Types of Data Analysis

Type	Key Question	What It Does	Example
Descriptive	What happened?	Identifies what has already occurred. First step before deeper analysis.	Google Analytics showing how many website visitors came last month
Diagnostic	Why did it happen?	Finds the reason behind an event. Like a doctor diagnosing a patient's illness.	Finding that retail clients leave faster than other clients due to lack of expertise

Type	Key Question	What It Does	Example
Predictive	What is likely to happen?	Uses historical data to forecast future trends. Machine learning makes this very accurate.	Insurance company predicting which drivers are more likely to have accidents
Prescriptive	What should we do?	Recommends the best course of action. Uses all other types combined. Most complex.	Google self-driving car algorithms making real-time driving decisions

Descriptive Analytics Techniques

- Data Aggregation: Gathering data and presenting it in a summarized format (e.g., total monthly sales)
- Data Mining: Exploring data to uncover hidden patterns or trends
- Tools: Google Analytics, HubSpot, Tableau, Infogram

Step 5: Sharing Your Results

What is it?
After analysis, you share insights with stakeholders (managers, decision-makers). This requires presenting data clearly and visually so everyone can understand.

Use reports, dashboards, and interactive visualizations to present findings.
The goal is to make insights 100% clear and unambiguous.
Remember: Visualization is great, but COMMUNICATION is the most important skill.
Tools (no coding needed): Google Charts, Tableau, Datawrapper, Infogram
Tools (with Python/R): Plotly, Seaborn, Matplotlib

Step 6: Embracing Failure

In data analytics, not all analyses give clear answers on the first try.
It is normal and expected to go back and revise your question, collect more data, or re-clean data.
Learning from failed or incomplete analyses makes you a better analyst.
Do not be afraid if results are unexpected - they may reveal something important.

Step 7: Summary of the Data Analysis Process

15. Define the Question: Set a clear, specific problem statement
16. Collect the Data: Gather first-party, second-party, or third-party data
17. Clean the Data: Remove errors, standardize, handle outliers and missing values

18. Analyze the Data: Use descriptive, diagnostic, predictive, or prescriptive methods
19. Share Your Results: Present with visualizations and clear communication
20. Embrace Failure: Revise and improve when results are unclear
21. Continuously improve your process with each project

Important Notes for Exam

- The data analysis process has 5 core steps (some textbooks list 7 with sub-steps)
- Data cleaning takes the MOST time in a real data project (up to 80%)
- The 4 types of analysis: Descriptive, Diagnostic, Predictive, Prescriptive
- Prescriptive is the most complex - it combines all other types
- Communication and visualization are just as important as technical analysis

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Chapter 4: Basic Statistics for Data Analytics

Why Statistics?
Statistics is the foundation of data analytics. It gives us the tools to summarize data (central tendency), understand how spread out the data is (dispersion), and display data clearly (frequency distribution and visualization).

4.1 Frequency Distribution

Definition
A frequency distribution is a representation - either in graphical or tabular format - that displays the number of observations within a given interval. Frequency = how often a value occurs. Distribution = the pattern of frequency of the variable.

It helps us see WHICH values appear most often in a dataset.
A frequency table lists all values and how many times each appears.
It can also be shown as a bar chart or histogram.

Example: 2022 Winter Olympics - Team USA Medals
Team USA won 25 medals at the 2022 Winter Olympics. The frequency table below shows the type and count:

Medal Type	Frequency (Count)
Gold	8

Medal Type	Frequency (Count)
Silver	10
Bronze	7
TOTAL	25

4.2 Data Presentation

Definition
Data presentation is the process of visually representing datasets to convey information effectively to an audience. It helps people understand complex data quickly.

Type	What It Is	When to Use
Graphical	Visual charts, graphs, diagrams	When you want to show trends, comparisons, or distributions visually
Numerical	Tables showing raw numbers	When exact values matter and comparison is needed

Key Graphical Tools

- Bar Diagram: Compares different categories using rectangular bars of different heights
- Pie Chart: Shows proportions of a whole as slices of a circle (percentages)
- Histogram: Similar to a bar chart but for continuous data grouped into intervals (bins). Note: Bar charts have gaps between bars; histograms do not.

4.3 Central Tendency

Definition
Central tendency is the statistical measure that identifies a single value as representative of an entire distribution. It is the single most typical value in a dataset. There are 3 measures: Mean, Median, and Mode.

Measure	Definition	How to Calculate	Example (Data: 2, 4, 4, 6, 8)
Mean (Average)	Sum of all values divided by count	Add all values, divide by total number	Mean = (2+4+4+6+8)/5 = 24/5 = 4.8
Median (Middle Value)	The middle value when data is sorted	Sort data, find the middle. If even count, average the two middle values.	Sorted: 2,4,4,6,8 - Middle = 4 (3rd value)

Measure	Definition	How to Calculate	Example (Data: 2, 4, 4, 6, 8)
Mode (Most Frequent)	The value that appears most often	Count frequency of each value, find the highest	4 appears twice - Mode = 4

Solved Example - Q1: Arithmetic Mean

Problem: Marks of 6 students: 20, 22, 24, 26, 28, 30. Find the Mean.

Solution:
Mean = (Sum of all values) / (Number of values)
Mean = (20 + 22 + 24 + 26 + 28 + 30) / 6
Mean = 150 / 6
Mean = 25

Answer: The mean mark is 25.

Solved Example - Q2: Finding Missing Observation

Problem: 14 observations: 26, 12, 14, 15, x, 17, 9, 11, 18, 16, 28, 20, 22, 8. Mean = 17. Find x.

Solution:
Step 1: Sum of known values = 26+12+14+15+17+9+11+18+16+28+20+22+8 = 216
Step 2: Mean formula: Mean = (Sum) / N => 17 = (216 + x) / 14
Step 3: 17 x 14 = 216 + x => 238 = 216 + x
Step 4: x = 238 - 216 = 22

Answer: The missing observation x = 22.

4.4 Dispersion (Measures of Spread)

Definition
Dispersion refers to how spread out or scattered the data values are around the average. It tells us whether the data is tightly packed around the mean or widely spread.

Type of Measure	Description
Absolute Measures	Use the same units as the original data. Examples: Range, Variance, Standard Deviation, Quartile Deviation, Mean Deviation.

Type of Measure	Description
Relative Measures	Unit-free; used to compare two or more datasets. Examples: Coefficient of Range, Coefficient of Variation, Coefficient of Standard Deviation.

Absolute Measures of Dispersion

Measure	Formula	Simple Explanation
Range	Range = X_max - X_min	Difference between the largest and smallest value. Simple but affected by outliers.
Variance (sigma^2)	Variance = SUM(X - mean)^2 / N	Average of squared differences from the mean. Shows overall spread.
Standard Deviation (sigma)	SD = sqrt(Variance)	Square root of variance. Same unit as data. Most commonly used.
Quartile Deviation	QD = (Q3 - Q1) / 2	Half the distance between the upper and lower quartiles. Good for skewed data.
Mean Deviation	MD = SUM|X - mean| / N	Average of absolute differences from the mean.

Relative Measures (Coefficients) of Dispersion

C.D. In Terms of	Formula
Range	C.D. = (X_max - X_min) / (X_max + X_min)
Quartile Deviation	C.D. = (Q3 - Q1) / (Q3 + Q1)
Standard Deviation	C.D. = SD / Mean (This is also called Coefficient of Variation)
Mean Deviation	C.D. = Mean Deviation / Average

Solved Example - Q3: Variance and Standard Deviation

Problem: Find Variance and Standard Deviation of: 1, 3, 5, 5, 6, 7, 9, 10

Step 1: Find the Mean
Mean = (1+3+5+5+6+7+9+10) / 8 = 46 / 8 = 5.75

Step 2: Subtract mean from each value (X - Mean)
(1-5.75)=-4.75, (3-5.75)=-2.75, (5-5.75)=-0.75, (5-5.75)=-0.75
(6-5.75)=0.25, (7-5.75)=1.25, (9-5.75)=3.25, (10-5.75)=4.25

Step 3: Square each difference (X - Mean)^2

22.563, 7.563, 0.563, 0.563, 0.063, 1.563, 10.563, 18.063

Step 4: Sum of squared differences = 61.504

Step 5: Variance = 61.504 / 8 = 7.69

Step 6: Standard Deviation = sqrt(7.69) = 2.77

Answer: Variance = 7.69, Standard Deviation = 2.77

Solved Example - Q4: Range and Coefficient of Range

Problem: Data: 45, 55, 63, 76, 67, 84, 75, 48, 62, 65

Step 1: Identify Maximum and Minimum values
X_max = 84, X_min = 45

Step 2: Calculate Range
Range = X_max - X_min = 84 - 45 = 39

Step 3: Calculate Coefficient of Range
C.D. = (X_max - X_min) / (X_max + X_min)
C.D. = (84 - 45) / (84 + 45)
C.D. = 39 / 129
C.D. = 0.302 (approximately)

Answer: Range = 39, Coefficient of Range = 0.302

Solved Example - Q5: Quartiles, IQR, and Median

Problem: Data: 19, 21, 23, 20, 23, 27, 25, 24, 31

Step 1: Arrange in ascending order
19, 20, 21, 23, 23, 24, 25, 27, 31 (n = 9 values)

Step 2: Find the Median (Q2)
Median position = (n+1)/2 = (9+1)/2 = 5th term
5th term = 23
Median = 23

Step 3: Find Lower Quartile (Q1)
Q1 position = (n+1)/4 = (9+1)/4 = 2.5th term
Average of 2nd and 3rd terms = (20+21)/2 = 20.5
Lower Quartile (Q1) = 20.5

Step 4: Find Upper Quartile (Q3)
Q3 position = 3(n+1)/4 = 3(9+1)/4 = 7.5th term
Average of 7th and 8th terms = (25+27)/2 = 26
Upper Quartile (Q3) = 26

Step 5: Inter-Quartile Range (IQR)
IQR = Q3 - Q1 = 26 - 20.5 = 5.5

Answer: Median = 23, Q1 = 20.5, Q3 = 26, IQR = 5.5

Important Notes for Exam

- Mean = Sum / Count. Best for symmetric data without outliers.
- Median = middle value. Best when data has outliers or is skewed.
- Mode = most frequent value. Can be used for categorical data too.
- Variance and Standard Deviation measure how spread out data is from the mean.
- Standard Deviation (SD) is more useful than Variance because it uses the same unit as the data.
- IQR = Q3 - Q1. It measures the spread of the middle 50% of data.
- A small SD means data is clustered close to the mean.
- A large SD means data is very spread out.

Chapter 5: Quick Reference Summary

Data Analysis Process - Summary Table

Step	Name	Key Action	Tools
1	Define Question	Identify the business problem (problem statement)	Grafana, Databox, KPI Dashboards
2	Collect Data	Gather first, second, or third-party data	Salesforce DMP, SAS, Pimcore
3	Clean Data	Remove errors, standardize, handle missing data	OpenRefine, Pandas (Python), Data Ladder
4	Analyze Data	Apply descriptive, diagnostic, predictive, or prescriptive analysis	Python, R, Excel, SPSS
5	Share Results	Visualize and present findings to stakeholders	Tableau, Google Charts, Matplotlib, Seaborn

4 Types of Data Analysis - Summary

Type	Question Answered	Bangladesh Example
Descriptive	What happened?	Pathao checks how many rides were completed last month
Diagnostic	Why did it happen?	Bkash finds out why mobile transactions dropped in a certain area
Predictive	What will happen?	10 Minute School predicts which students are likely to drop out
Prescriptive	What should we do?	An e-commerce site recommends what discount to offer to retain a customer

Statistics Formulas Quick Reference

Formula	Expression	Use When
Mean	Mean = Sum of all values / N	Finding average of a dataset
Variance	Variance = SUM(X - Mean)^2 / N	Measuring how spread out data is
Standard Deviation	SD = sqrt(Variance)	Measuring spread in the same unit as data
Range	Range = X_max - X_min	Finding the total spread of data
Coefficient of Range	C.D. = (X_max - X_min) / (X_max + X_min)	Comparing spread of two datasets
Median	(n+1)/2 th term (if sorted)	Finding the middle value
Lower Quartile Q1	(n+1)/4 th term	Finding the 25th percentile
Upper Quartile Q3	3(n+1)/4 th term	Finding the 75th percentile
IQR	IQR = Q3 - Q1	Measuring spread of middle 50% of data

Important Notes for Exam

- Data Quality has 6 dimensions: Accuracy, Completeness, Consistency, Validity, Uniformity, Relevance
- Three types of data: First-party (own), Second-party (partner's), Third-party (external/public)
- Data cleaning takes up 80% of a data analyst's time in practice
- An outlier should ONLY be removed if it can be proven to be an error
- Prescriptive analytics is the most advanced - it combines all other types
- Visualization tools without coding: Tableau, Google Charts, Datawrapper, Infogram
- Python libraries for visualization: Matplotlib, Seaborn, Plotly
- Standard Deviation is preferred over Variance because it uses the same unit as the data

--- End of Notes ---